SELF-REFLECTIVE REPORT: OBJECT ORIENTED PROGRAMMING BIT1123 / ASSIGNMENT 1

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1. INTRODUCTION

Object-Oriented Programming (OOP) is one of the key software programming paradigms. During the BIT1123 Object-Oriented Programming module at City University Malaysia, I attended a structured weekly lab tutorial, where I learned the fundamental programming concepts of Java. The path of my technical learning, challenges encountered, solutions implemented, and development and version control growth in Java is reflected in this report.

2. The knowledge acquired from each tutorial is:

Week 1: Basic syntax, data types, console I/O and simple decision-making logic with arithmetic.
- Week 2: Making own classes, creating instances of own classes using constructors, calling methods on instances.
- Week 3-4: Application of control flow statements (if-else, loops), array operations and multi-class relationships.
- Week 5: Data hiding and encapsulation through private fields, public getters, setters, constructors and proper documentation.
Inheritance, method reusability and constructor referencing of the superclass (`super`) were covered in the class hierarchy design of this week (Week 6).
Week 7: Dynamic Polymorphism, overriding methods, run-time binding, manipulating object behaviour.
Week 8-9: Abstract Classes and Interfaces – architectural abstraction and design contracts.
Week 10: Event driven programming and graphical user interface (GUI) development using Java Swing/AWT components.

3. CHALLENGES ENCOUNTERED :

As I worked on the practical tutorials, there were a few technical challenges I faced:
Understanding Object References vs. Primitive Types: Knowing the difference between how the memory works and equality checking (`==` versus `.equals()`).
Bugs: Inheritance and Polymorphism: Rules and type casting.
Learners will master the principles of Git version control, including how to use the command line, the structure of repositories and working with project synchronization on GitHub.

4. How the Challenges were overcome :

Hands-on Debugging: Used the debugging tools of IntelliJ IDEA to follow the states of objects and the allocation of memory.
A. Code Refactoring - Tutorial exercises updated on a regular basis using Java documentation and feedback from faculty.
- Version Control Practice: Used a structured Git work-flow, such as making commits, pushing, organizing folders etc., within GitHub directly.

5. Improvements in Java programming skills :

My programming skills progressed from creating simple sequential programs to development of structured, maintainable, and modular Java programming throughout the semester. I learned to use the right access modifiers, package organization and using the IDE shortcuts to help with debugging.

A knowledge of the concepts of object-oriented programming.
I got a thorough understanding of the four pillars of OOP:
- Encapsulation: Access control to protect the internal states of an object.
Inheritance: Making code reusability by creating relationships between classes called parent-child.
Polymorphism: OoD using a single interface to represent multiple underlying forms.
Hiding implementation details in well-defined interfaces.

7. FUTURE LEARNING PLANS

In the future, I would like to learn more about advanced Java concepts such as Exception Handling, Data Structures, Multithreading, and Database Connectivity (JDBC). Also, I plan on using GitHub for a lot of times to create personal portfolios as well as open-source contributions.

8. CONCLUSION & REPOSITORY LINK

The practical sessions really helped a lot in reinforcing the theoretical part of my Object Oriented Programming. The addition of tutorial work into a single, centralized Git repository necessitated industry standard practices.

GitHub Repository URL: https://github.com/saiffonee-bot/OOP

- **APPENDIX: PRACTICAL SCREENSHOTS**

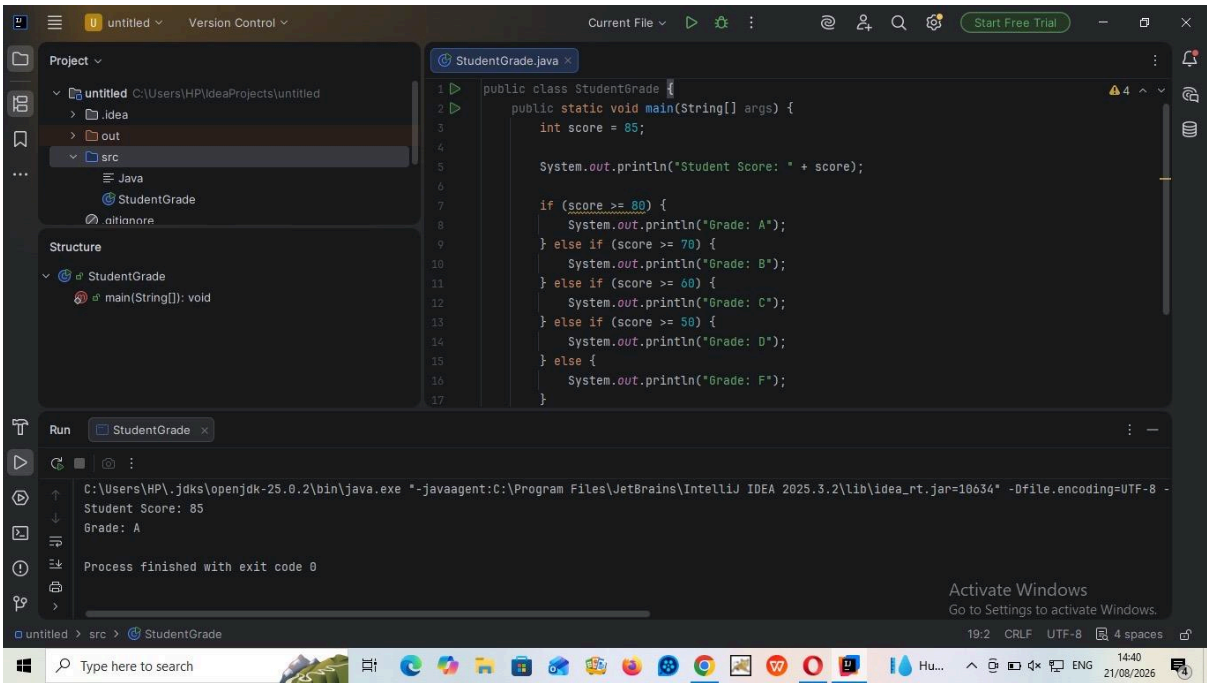

*Figure 1: Successful execution of Java code (StudentGrade.java) inside IntelliJ IDEA .

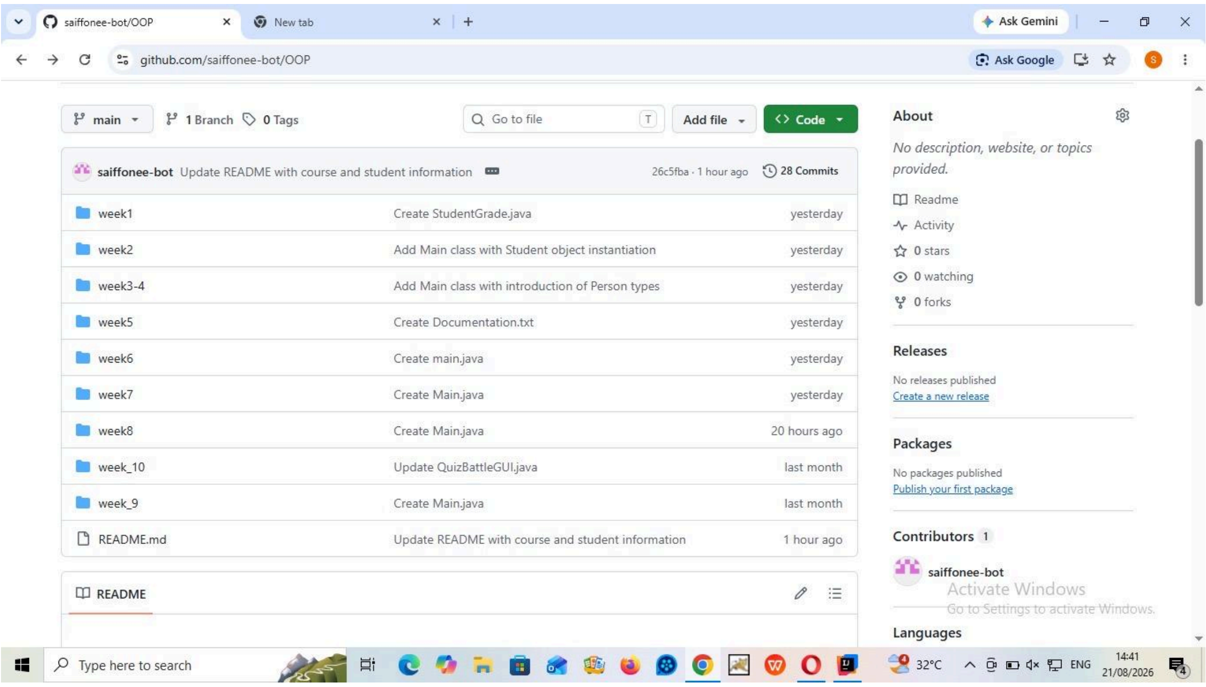

*Figure 2: Structure of the GitHub repository containing weekly OOP learning log and tasks.